

de Rham Cohomology

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The only thing we truly understand in mathematics is linear algebra, and all of mathematics is about reducing difficult questions to linear algebra.

— Folklore Saying

1 Introduction

de Rham cohomology deals with understanding the topological properties of a space by doing calculus on it. The motivation and first examples of the theory come from vector calculus. In $\mathbb{R}^2$, we can take the gradient of a function to get a vector field $f \mapsto (f_x, f_y)$ and take the "curl" $(f, g) \mapsto f_y - g_x$. A simple calculation shows that the composition of these two operations is always zero. A fact that I always found surprising was that a condition for a vector field F to be conservative i.e. $F = \nabla f$ for some f , a sufficient condition is to have a vanishing curl i.e. $\nabla \times F = 0$. This is an important problem e.g. in physics, f is a "potential function" to a vector field (gravitational, electric), which makes computations easier (in particular, there is no "magnetic potential" as the curl of the magnetic field is nonzero).

The idea for proof is simply to construct a primitive! Fix some point a and for any other point x , let γ be a path connecting a to x and define $f(x) = \int_{\gamma} F dx$. By the fundamental theorem of (multivariable) calculus, we have $F = \nabla f$. Why doesn't this work for any vector field? The value of f depends a priori on the path γ , but in fact, it can be shown that for a vector field with vanishing curl, the integral is independent of the path, so that f is well defined. The main point is to use Stoke's theorem to show that line integrals of F are path independent and hence you can integrate it to get a well defined "potential function" $f = \int F \cdot dr$, which by the fundamental theorem of calculus implies $F = \nabla f$. This has the surprising consequence that for a vector field to be the gradient of a potential, it is sufficient for it to be "irrotational," which at first seems completely unrelated!

This is a great theorem! Unfortunately, it is false. Consider the vector field

$$F = \left(\frac{y}{x^2 + y^2}, -\frac{x}{x^2 + y^2} \right),$$

which is the vector field of unit vectors pointing counter-clockwise. A direct calculation would show the curl of this vector field is 0. However, the vector field cannot be the gradient of a function.

Suppose we use the same method as before and define $f(x, y) = \int F \cdot dr$ on any path starting from the point $(1, 0)$ to (x, y) . However, this function is not well defined because $f(1, 0)$ should be 0 using the constant path, but $f(1, 0) = 2\pi$ by integrating over the counter-clockwise unit circle. (In fact, any two circles of arbitrary radius are "homologous" i.e. together bound a surface (an annulus), so by the argument above, give the same integral as $\nabla \times F = 0$). You can try to fix this by restricting the paths you choose so that each endpoint has only one possible value (e.g. exclude the unit circle path), however you can convince yourself that f will not be continuous everywhere, and hence not differentiable everywhere. To see this a bit clearer, formally, $F = \nabla \arctan(y/x)$, but this is undefined at $x = 0$ and you can find a different primitive, but they will all be undefined at some point.

The reason for this counterexample is subtle: F is not defined at 0. This seemingly insignificant fact actually has deep implications. First, why does the proof fail? Stoke's theorem requires the interior of a loop to be simply connected, however F is defined on $\mathbb{R}^2 - \{0\}$ so any curve containing the origin (such as the unit circle) has an interior which is not simply connected and hence, the line integral over such a loop is not necessarily zero. What is maybe more surprising is that we will see that in a precise sense, this vector field is the "only obstruction" to a vector field defined over $\mathbb{R}^2 - \{0\}$ being the gradient of a function. More precisely, if G is another vector field that is defined over $\mathbb{R}^2 - \{0\}$ but cannot be expressed as the gradient of a function, there is some $\lambda \in \mathbb{R}$ such that $G - \lambda F$ is the gradient of some function. In other words, we can always remove the "non-conservativeness" using F . The value of λ is also easy to calculate: $\lambda = \frac{1}{2\pi} \int_{\gamma} G \cdot dr$ where γ is the counter-clockwise unit circle. In a similar fashion, suppose H is a vector field on $\mathbb{R}^2 - \{(0,0), (a,b)\}$ with zero curl but not conservative. Define the vector field G as $G(x,y) = F(x-a, y-b)$ i.e. the counter-clockwise swirl centered at (a,b) . Then, there exist $\lambda, \mu \in \mathbb{R}$ such that $H - \lambda F - \mu G$ is the gradient of a function. The existence of two vector fields to correct the failure corresponds to the two holes in the space. The main idea is that we know $\nabla \times \nabla$ is 0 on every function and when we ask the reverse question (when can a vector field which has 0 curl be expressed as the gradient of a function), the failure of a positive answer gives information about the topology.

In 3-dimensions, we have 3 differential operators, namely the gradient ∇ , the curl $\nabla \times$, and the divergence $\nabla \cdot$. From a direct computation, the composition of two of these operators is identically zero. I always remembered these with the "sequence"

$$\text{functions} \xrightarrow{\nabla} \text{vector fields} \xrightarrow{\nabla \times} \text{vector fields} \xrightarrow{\nabla \cdot} \text{functions}.$$

As we'll see later, this turns out to be very important! At the middle two vector fields parts of the sequence, the composition of the map going out and the map going in is 0. We can ask the reverse question at each step again. First, if a vector field has 0 curl, is it necessarily the gradient of a function? As a counterexample, we can extend our 2D case: $F = \left(\frac{y}{x^2+y^2}, -\frac{x}{x^2+y^2}, 0 \right)$. Here, the vector field is undefined on the z -axis so this vector field detects a "1-dimensional hole."

Now, we can look at the second part. If a vector field has 0 divergence, is it the curl of another vector field? Again, the answer is (surprisingly, with proof omitted) yes if there are no holes, but no in general. The counter example is the vector field

$$F = \left(\frac{x}{(x^2 + y^2 + z^2)^{3/2}}, \frac{y}{(x^2 + y^2 + z^2)^{3/2}}, \frac{z}{(x^2 + y^2 + z^2)^{3/2}} \right).$$

A direct calculation will show $\nabla \cdot F = 0$. We want to show that this field cannot be the curl of another vector field. This is more difficult than the last question, because as we know how to produce a primitive, it is difficult to produce an "explicit inverse curl." Nevertheless, we can be clever and use an indirect argument. Let S^2 be the unit sphere at the origin and consider the integral $\int_{S^2} F \cdot dn$ where dn denotes the unit outward facing normal. It can be checked that this evaluates to 4π . This forms a counterexample to the divergence theorem similar to how the last problem was a counterexample of Stoke's theorem. The divergence theorem would imply

$$\int_{S^2} F \cdot dn = \int_V \nabla \cdot F dV = 0$$

since the divergence is 0. But this is not true, and again, the issue is caused by the vector field being undefined at zero (although this isn't enough to guarantee a failure, for example $\nabla \times ((x^2 + y^2 + z^2)^{-1}, 1, 1)$ isn't defined at the origin, but still obeys the divergence theorem). How can we use this to show that $F \neq \nabla \times G$ for some other vector field G ? We can exploit Stoke's theorem. If the statement were true, we could take S to be the upper hemisphere of the unit sphere and ∂S is the unit circle oriented counterclockwise. Then, Stoke's theorem would tell us

$$\int_{\partial S} G \cdot dr = \int_S \nabla \times (G) \cdot dn = \int_S F \cdot dn$$

where dn points upward so that it is positively oriented (outward of the sphere). Similarly, if we let S' denote the lower hemisphere and $\partial S'$ denote the boundary, we have also

$$\int_{\partial S} G \cdot dr = \int_S \nabla \times (G) \cdot dn = \int_{S'} F \cdot dn$$

where the normal vector points upward again (inward on the sphere). Hence, we should have

$$\int_S F \cdot dn - \int_{S'} F \cdot dn = \int_{\partial S} G \cdot dr - \int_{\partial S} G \cdot dr = 0.$$

But we also have

$$\int_S F \cdot dn - \int_{S'} F \cdot dn = \int_{S^2} F \cdot dn$$

(the negative sign to switch the orientation to outward of the sphere), and we showed this value was 4π . Hence, we have a contradiction and we cannot have $F = \nabla \times G$. Here, F is undefined at the origin, so we detect a "0-dimensional hole."

We can notice some commonalities. In the 2 dimensional case, our failure measured 0-dimensional holes, while in the 3 dimensional case, our failure at the first step measured 1-dimensional holes and our failure at the second step measured 0-dimensional holes. The first step in both cases is the functions which have 0 curl but aren't the gradient of another function, but the first measures 0-dimensional holes while the second measures 1-dimensional holes. We can unify both of these into saying that they measure codimension-2 holes. Now, the second step in the 3 dimensional case measures codimension-3 holes. It makes sense why we don't have a "second step" in two dimensions because we can't have a codimension 3 object in the plane.

Now, you should really want to find a piece that measures codimension-1 holes. The only place this can go is to look at the functions which have 0 gradient but aren't the image of... well there's nothing to compare it to, so let's just study functions which have 0 gradient everywhere. You may immediately think that the solutions are all constant functions, but there is a bit of a technicality. To discuss this, let's look at examples of codimension 1-holes. In 2-dimensions, we can take a line which splits the plane into two parts, or take the unit circle which splits the plane into the interior and exterior. Similarly, in three dimensions, we can split $\mathbb{R}^3$ by cutting it using a plane, or a unit sphere, or a paraboloid, or any other surface. Now, we notice that functions on these spaces with zero gradient aren't everywhere constant functions, but functions which are constant on each connected component. For example, if we split the plane by the x -axis, we can choose a function which is 0 on the lower half plane and 1 on the upper half plane. In general, the dimension of the vector space which has 0 gradient is the number of connected components of the space. You may argue that this is not the same as "codimension-1 holes," but if you look at examples of cutting the plane by the x and y axis, or the plane cut by both of these and the unit circle, I suggest you to find a nice definition of "codimension-1 holes" which is well behaved under deformations of the space. You will find it difficult to find a good definition.

Our "codimension-k holes" is a good intuition, but it isn't exactly what's happening. The codimension-2 holes some hole which prevents you from continuously deforming an S^1 to a point. For example, in the plane without the origin, the unit circle cannot be deformed to a point because at some point in the deformation, a point on the circle would have to pass through the origin. Verify that the codimension-2 hole in $\mathbb{R}^3$ does the same. Now, the codimension-3 hole in $\mathbb{R}^3$ represents an obstruction to deforming a sphere S^2 to a point. For example, the hole at the origin prevents us from deforming the unit sphere to the origin. This is strongly related to the failure of the functions to satisfy Stoke's and the divergence theorems. Similarly, our description of codimension-1 holes is what prevents us from deforming S^0 to a point. Since $S^n = \partial D^{n+1}$, we see S^0 is a pair of points. A pair of points can be deformed to a point if and only if both points lie in the same connected component. Hence, this gives a better picture of what's actually being represented. It is not clear why this data of functions is related to how S^k can't be compressed to a point. de Rham's original work was in fact showing this isomorphism between the de Rham cohomology which we will study, and the singular homology, which is the data

of "failure of S^k to be contracted to a point." This statement is beyond the notes here, but because this isomorphism exists, we will freely utilize the intuition on the topology of our space to predict answers to these problems involving differential operators.

2 The Cochain Complex

In this section we will properly understand each part of the sequence

$$\text{functions} \xrightarrow{\nabla} \text{vector fields} \xrightarrow{\nabla \times} \text{vector fields} \xrightarrow{\nabla \cdot} \text{functions}.$$

There are two parts which must be addressed: what is the generalization of functions and vector spaces, and what is the generalization of the differential maps. The gradient going from functions to vector spaces and the divergence going from vector spaces to functions are easily generalizable. The strange part is the curl, which requires the cross product.

2.1 Wedge Products

The cross product outputs a vector that's orthogonal to two given vectors, but it's secretly representing the plane that contains the two vectors. This trick only works in three dimensions because there is a unique normal vector to every plane. To give some more evidence as to why the curl should output a 2-dimensional object, the "curl" formula works in $\mathbb{R}^2$ which is sometimes thought of as the z -component of the 3D curl. But it's inconvenient to include an unnecessary third dimension, so it's better to think of it as an intrinsic object. If we replace the derivatives with vector components, it becomes a determinant of two vectors (similar to how the 3D curl becomes a cross product). The determinant gives the area spanned by two vectors suggesting we should look for some sort of "area element."

A vector can be described as something with a magnitude and direction. Alternatively, we can say it has a length and an orientation. The two dimensional generalization we seek is something with an area and an orientation (the orientation can be thought of as the sign of the determinant being positive if it's oriented ccw and negative if cw). A very natural object to describe an area is a parallelogram spanned by two vectors v, w , which we will identify with the vectors themselves. Hence, we write our "area element" as (v, w) . There are multiple pairs of vectors which give the same area, and hence should be identified together. Taking inspiration from geometry, if we shear a parallelogram, the area does not change. Hence, we should identify the pair (u, v) with $(u + \lambda v, v)$ and $(u, v + \lambda u)$ for $\lambda \in \mathbb{R}$ (corresponding to shearing along the first and second edge respectively). We also have that scaling either edge by some factor also scales the area by that factor. Hence, we should identify $(\lambda u, v)$, $(u, \lambda v)$, and $\lambda(u, v)$ as the same element (the multiplication by λ in the last term is just formal multiplication for now). It can be verified that if we declare (v_1, v_2) to have area 1 if v_1, v_2 are positively oriented and orthonormal and $\lambda(u, v)$ has area λ times the area of (u, v) , then these two axioms are enough to determine the area of any parallelogram. This argument works for any $\mathbb{R}^n$ with $n \geq 2$ and here, two area elements are identified if they span the same plane (the orientation) and have the same (signed) area.

Exercise (Equivalence of Axioms)

For $\lambda, \mu \in \mathbb{R}$ and vectors u, v, w , show that the axioms

1. (Shearing) $(u, v) = (u + \lambda v, v) = (u, v + \lambda u)$
2. (Scaling) $(\lambda u, v) = (u, \lambda v) = \lambda(u, v)$

is equivalent to the following axioms

1. (bilinearity) $(\lambda u + \mu v, w) = \lambda(u, w) + \mu(v, w)$ and $(u, \lambda v + \mu w) = \lambda(u, v) + \mu(u, w)$

2. (anticommutativity) $(u, v) = -(v, u)$

Hence, we can identify "area elements" by pairs of vectors that satisfies the bilinearity and anticommutativity axioms. This motivates the following definition:

Definition 1 (Wedge Product). *Given a vector space V , we define the vector space $V \wedge V$ as being spanned by symbols of the form $v \wedge w$ for $v, w \in V$ satisfying the following axioms:*

1. (bilinearity) $(\lambda u + \mu v) \wedge w = \lambda u \wedge w + \mu v \wedge w$ and $u \wedge (\lambda v + \mu w) = \lambda u \wedge v + \mu u \wedge w$
2. (anticommutativity) $u \wedge v = -v \wedge u$

Although it doesn't have any geometric significance, $V \wedge V$ can have elements which are formal sums of "areas" not lying in the same plane. For example, for $V = \mathbb{R}^4$ with basis e_1, e_2, e_3, e_4 , $e_1 \wedge e_2 + e_3 \wedge e_4 \in V \wedge V$, but cannot be expressed as $v \wedge w$ for $v, w \in V$. This is in fact the smallest nontrivial example of such phenomenon as it turns out that if $V = \mathbb{R}^2$ $V \wedge V$ is one dimensional, and if $V = \mathbb{R}^3$, every element can be expressed as $v \wedge w$ for $v, w \in V$, reflecting the fact that there is a unique direction orthogonal to every plane (exercise).

The reason we allow these formal combinations is so that $V \wedge V$ is a vector space and we can use linear algebra. Although the geometric axioms of shearing and scaling are more intuitive, the algebraic axioms of bilinearity and anticommutativity are more convenient for the linear algebra. Note that $v \wedge v = 0$ (why?) so that $\mathbb{R}^1 \wedge \mathbb{R}^1$ is 0 dimensional (representing there's no areas). We can easily generalize this concept to represent n -dimensional volumes.

Definition 2 (n -th Wedge Product). *For V a vector space, define $\bigwedge^1 V = V$, $\bigwedge^2 V = V \wedge V$ and $\bigwedge^n V = (\bigwedge^{n-1} V) \wedge V$. More concretely, $\bigwedge^n V$ is spanned by symbols of the form $v_1 \wedge v_2 \wedge \dots \wedge v_n$ satisfying the following axioms:*

1. (multilinearity) $v_1 \wedge \dots \wedge (\lambda u + \mu w) \wedge \dots \wedge v_n = \lambda v_1 \wedge \dots \wedge u \wedge \dots \wedge v_n + \mu v_1 \wedge \dots \wedge w \wedge \dots \wedge v_n$, where the $\lambda u + \mu w$ takes the place of any v_i
2. (anticommutativity) $v_1 \wedge \dots \wedge v_i \wedge \dots \wedge v_j \wedge \dots \wedge v_n = -v_1 \wedge \dots \wedge v_j \wedge \dots \wedge v_i \wedge \dots \wedge v_n$ for every $i \neq j$, or alternatively, given $\sigma \in S_n$, $v_1 \wedge \dots \wedge v_n = (-1)^{\text{sgn}(\sigma)} v_{\sigma(1)} \wedge \dots \wedge v_{\sigma(n)}$

Here, an element of $\bigwedge^n V$ is a formal combination of n -fold wedge products which represent n -dimensional volumes. It can be checked that similar to the two-dimensional case, we can describe n -dimensional volumes using the shearing and scaling axioms in each component, and these are equivalent to the axioms above.

Exercise (Dimension of Wedge Product Spaces)

If $V = \mathbb{R}^n$, $\dim(\bigwedge^k V) = 0$ for $k > n$. Further show that for $0 \leq k \leq n$, $\bigwedge^k V$ is spanned by $v_{i_1} \wedge \dots \wedge v_{i_k}$ where v_i forms a basis of V and $i_1 < i_2 < \dots < i_k$. Hence show $\dim(\bigwedge^k V) = \binom{n}{k}$. Based on the dimension count, $\bigwedge^n V$ is one-dimensional, while $\bigwedge^{n-1} V$ is n -dimensional and so these are sometimes called pseudoscalars and pseudovectors due to this "duality" with scalars and vectors.

Now really, this is a bit awkward since we're taking formal linear combinations of areas. Also, this tells us when two areas are equivalent, but it does not actually give us a value for the area of these parallelograms. In order to remedy both of these issues, we can study the dual space

Definition 3. *Dual Space For a $\mathbb{R}$ vector space V , the dual space V^* is defined as the vector space $\text{Hom}(V, \mathbb{R})$. In other words, it is the space of linear functions from V to $\mathbb{R}$.*

If V is thought of as column vectors, V^* can be thought of as row vectors whose multiplication with the column vector gives a real number. Formalizing this shows that the dimension of V^* equals the dimension of V (exercise).

Now, consider the following scenario: you want to compute the area of the parallelogram spanned by two vectors in three dimensional space. By shearing and scaling, you can get an equivalence of parallelograms of the same area, but you can't actually compute the area. However, if you specify a square of area one, then you can check how much this parallelogram is a multiple of the unit square and that specifies an area. Now, one idea is to take a "basis" of parallelopipeds and specify the area of these and then extend by linearity to the entire space (since we still want to maintain some form of linearity). This is equivalent to the data of a linear function of map from $\bigwedge^k(V)$ to $\mathbb{R}$ i.e. an element of $\left(\bigwedge^k V\right)^*$.

Exercise (Wedge of the Dual is the Dual of the Wedge)

Show that $\bigwedge^k(V^*) = \left(\bigwedge^k V\right)^*$. The main idea is that given k dual vectors, we can feed in k vectors and multiply them to get out a real number. This number is supposed to represent some form of area of the parallelopiped spanned by these vectors. However, in order for it to represent an area, it should be antisymmetric in its arguments (it's already linear) so as to respect the shearing and scaling axioms. In order to force antisymmetry, we can use an "averaging trick":

$$(\omega_1, \dots, \omega_k)(v_1, \dots, v_k) = \sum_{\sigma \in S_n} (-1)^{\text{sgn}(\sigma)} \prod_{i=1}^k \omega_i(v_{\sigma(i)}).$$

Verify that such functions lie in, and form a basis of $\left(\bigwedge^k V\right)^*$. It may be worth expanding the right side in simple cases to see the analogy with the determinant! For every element of $\omega \in \bigwedge^k(V^*)$, we now have an object that actually assigns an area to every k set of vectors v_i by taking $\omega(v_1, \dots, v_k)$, and we can say if two sets are equivalent if they span the same subspace and have the same output of this function. This also fixes the problem of taking formal linear combinations of areas - now it's just adding functions, which feels more natural.

2.2 Differential Forms

We are finally in a position to generalize our sequences of differential operators. First, we need to generalize the spaces of functions and vector fields. In our 3D example, the second "vector fields" secretly represents areas and the second "functions" secretly represents volumes as we shall see (pseudovectors and pseudoscalars). For $\mathbb{R}^n$ spanned by $x_1, \dots, x_n$, we will work with the formal symbols $dx_1, \dots, dx_n$ called differentials which can be thought of as "infinitesimal lengths." The generalizations of the spaces will be seen to be "infinitesimal k -volumes."

Definition 4 (Differential Forms). Let V^* be the vector space spanned by $dx_1, \dots, dx_n$. We define the space of differential k -forms on $X \subset \mathbb{R}^n$ to be $\Omega^k(X) = C^\infty \otimes \bigwedge^k V^*$. More concretely, elements of $\Omega^k(X)$ are formal sums of the form $\sum_I f_I dx_I$ where $I = (i_1, \dots, i_k)$ is a multi-index and $dx_I = dx_{i_1} \wedge \dots \wedge dx_{i_k}$.

dx_i essentially assigns a dual vector to every point: at every point it takes in a vector and returns the x_i component, or alternatively said, it takes a vector field as input and returns a C^∞ function f where $f(p)$ is obtained by taking the x_i 'th component of the vector at p . Thus, $\Omega^k(X)$ eats k vectors at every point and returns a real number giving a notion of the "infinitesimal area" spanned by these functions. Note in general (if X is not a subspace of $\mathbb{R}^n$), $\Omega^k(X)$ should be replaced with the space of sections of the k -th wedge power of the cotangent bundle, which forms a coordinate independent way of thinking about it.

Some notational conventions: $\Omega^k(X)$ will be denoted Ω^k when there's only one space and the wedges in $dx_{i_1} \wedge \dots \wedge dx_{i_k}$ are usually omitted so it's written as $dx_{i_1} dx_{i_2} \dots dx_{i_k}$, keeping in mind $dx_i dx_j = -dx_j dx_i$ to maintain the wedge product axioms. Later, this definition will be generalized to consider

more general topological spaces for X . To get a better idea for these spaces, let's study them when $X = \mathbb{R}^3$.

$$\Omega^0(\mathbb{R}^3) : f$$

$$\Omega^1(\mathbb{R}^3) : f dx + g dy + h dz$$

$$\Omega^2(\mathbb{R}^3) : f dy dz + g dz dx + h dx dy$$

$$\Omega^3(\mathbb{R}^3) : f dx dy dz$$

for $f, g, h \in C^\infty(\mathbb{R}^3)$. These are the actual spaces that are in our sequence with functions and "vector fields." It should be clear how to identify 1-forms and 2-forms as vector fields and why we often treat 3-forms simply as functions. We should think of elements of Ω^k as "infinitesimal volume determining functions" against which we can integrate.

2.3 The Exterior Derivative

We now look at how to generalize our maps between these spaces. Given a function $f \in C^\infty(X)$, we can look at the symbol df which intuitively is an infinitesimal change in f . We define it as $df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$ and $df_p(v)$ (the dual vector at the point p evaluated at the vector v at p) represents the idea of how much f changes if we move in the direction v at p . We generalize this to maps $\Omega^k \rightarrow \Omega^{k+1}$.

Definition 5 (Exterior Derivative). *We define a map called the exterior derivative $d^k : \Omega^k \rightarrow \Omega^{k+1}$ as follows: given $F = \sum f_I dx_I \in \Omega^k$, define $d^k F = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i dx_I$ (the differentials are multiplied by taking dx_i first and then dx_I in that order).*

The superscript on d is often omitted when it's clear from context. Let's see how these maps look like in $X = \mathbb{R}^3$:

$$d^0 : dF = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$d^1 : F = f dx + g dy + h dz$$

$$\begin{aligned} dF &= \frac{\partial f}{\partial x} dx dx + \frac{\partial f}{\partial y} dy dx + \frac{\partial f}{\partial z} dz dx + \frac{\partial g}{\partial x} dx dy + \frac{\partial g}{\partial y} dy dy + \frac{\partial g}{\partial z} dz dy + \frac{\partial h}{\partial x} dx dz + \frac{\partial h}{\partial y} dy dz + \frac{\partial h}{\partial z} dz dz \\ &= \left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z}\right) dy dz + \left(\frac{\partial f}{\partial z} - \frac{\partial h}{\partial x}\right) dz dx + \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dx dy \end{aligned}$$

$$d^2 : F = f dy dz + g dz dx + h dx dy$$

$$dF = \frac{\partial f}{\partial x} dx dy dz + \frac{\partial g}{\partial y} dy dz dx + \frac{\partial h}{\partial z} dz dx dy = \left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}\right) dx dy dz$$

where we used $dx_i dx_i = 0$ and changed signs when commuting $dx_i dx_j = -dx_j dx_i$. Notice that d^0 is the gradient, d^1 is the curl, and d^2 is the divergence! Hence, this seems to be the right generalization of our sequence which is clear to define for any dimension. We have finally arrived at the following:

Definition 6 (Differential Cochain Complex). *The differential cochain complex is defined as the following spaces and maps for X a subset of $\mathbb{R}^n$:*

$$0 \rightarrow \Omega^0 \xrightarrow{d^0} \Omega^1 \rightarrow \dots \rightarrow \Omega^{n-1} \xrightarrow{d^{n-1}} \Omega^n \xrightarrow{d^n} 0.$$

The first and last spaces are the 0-vector space where the first map is the injection of 0 (sometimes denoted d^{-1} for convenience) and the last map is justified since there are no $n+1$ forms (we computed the dimension of the space of degree k wedge product in an exercise, and this implies there are none for $k > n$). Our earlier comments on the topology of a space being determined by the calculus says that holes are determined by the space $\ker(d^k)/\text{im}(d^{k-1})$.

Definition 7 (Cohomology Spaces). The vector spaces $H^k(X)$ called the k -th cohomology space is defined as $\ker(d^k)/\text{im}(d^{k-1})$.

These objects will be our main objects of study going forward. Elements of $\ker(d)$ will be known as closed forms and elements of $\text{im}(d)$ will be known as exact forms.

Exercise ($d^2 = 0$)

If we write an expression like $d \circ d(f)$, for a k -form f , what are the indices for each d ? Show that for any k -form f , $d^2 f = d \circ d(f) = 0$.

We have now discussed how to construct our differential cochain complex

$$0 \rightarrow \Omega^0(X) \xrightarrow{d} \Omega^1(X) \rightarrow \dots \rightarrow \Omega^{n-1}(X) \xrightarrow{d} \Omega^n(X) \xrightarrow{d} 0.$$

by defining differential k -forms and the exterior derivative d for any space $X \subset \mathbb{R}^n$. However, there are many topological spaces that we are used to which we would also like to understand the topology of such as spheres S^n and tori $T^n = (S^1)^n$. Our main goal with de Rham cohomology is to understand the topology in terms of the calculus on the surface. It makes intuitive sense to do calculus on a sphere for example: imagine a heat map on S^2 , that is a function which assigns to every point a real number in a "smooth" manner, then it makes intuitive sense that the gradient should be something that points toward the greatest amount of change in this function. However, if we try to define the derivative $f'(x) = \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x}$, it doesn't make sense on a sphere because there is no linear structure on the sphere to take the difference $y - x$. You could try to embed it into $\mathbb{R}^3$ and take the gradient, but now the gradient is some vector in $\mathbb{R}^3$ and is not intrinsic to S^2 . Instead, we try to formulate a language on which we will be able to do calculus on such objects, and use it to understand the topology.

3 Manifolds

One way that we can try doing calculus is by deforming the surface into something we know how to do calculus in, in particular $\mathbb{R}^n$. The calculus we want to do is some generalization of the derivative, but the derivative is a very local property so we can just try to do calculus in the vicinity of every point and patch it together at the end. These two steps will lead us very naturally to the definition of a manifold.

Consider the upper hemisphere of the sphere S^2 . We can take a homeomorphism φ of this hemisphere, which we'll call U to some subset $\varphi(U)$ of $\mathbb{R}^2$. Given a function $f : S^2 \rightarrow \mathbb{R}$, we get an induced function $f \circ \varphi^{-1} : U \rightarrow \mathbb{R}$. Here we see that it is important for φ to be a homeomorphism so that φ^{-1} exists. Since we know how to do calculus on $\mathbb{R}^2$, we could define continuity and differentiability of f by declaring that $f \circ \varphi^{-1}$ is continuous or differentiable on $\varphi(U)$. Hence, we can transfer concepts of calculus from $\mathbb{R}^2$ to S^2 . There is a small technicality: continuity and differentiability requires limits in $\mathbb{R}^2$ coming from every direction. Hence, for any point $p \in U$, we want some set of points in every direction of its vicinity to be in $\varphi(U)$. In other words, $\varphi(U)$ must contain some small open ball around $\varphi(p)$ in $\mathbb{R}^2$ (imagine we took U to be the closed hemisphere, then for a point p on the equator, we could only take limits around $\varphi(p)$ in some directions and hence can't properly do limits). Equivalently, we declare that $\varphi(U)$ must be open and hence, since preimages of open sets are open, U is an open subset of S^2 .

We want to also make sure that we can do this procedure of locally transferring to $\mathbb{R}^n$ and doing calculus at every point. Hence we reach our first axiom of manifolds: a manifold M must be covered by open sets U_i with corresponding homeomorphisms (called chart maps) $\varphi_i : U_i \rightarrow \mathbb{R}^n$ for some n such that $M = \bigcup U_i$. The first part says that we want to be able to locally understand patches of M by converting it to $\mathbb{R}^n$ while the second part says that we can do this at every point.

Now there's a bit of ambiguity: the U_i and φ_i can be chosen differently so that there are multiple different ways of doing calculus around a point p . Suppose on a manifold M , we have $p \in U_1 \cap U_2$

and we have chart maps φ_1 and φ_2 on U_1 and U_2 . Then, we can describe the calculus of a function $f : M \rightarrow \mathbb{R}$ in the vicinity of p by $g_1 = f \circ \varphi_1^{-1} : V_1 \rightarrow \mathbb{R}$ and $g_2 = f \circ \varphi_2^{-1} : V_2 \rightarrow \mathbb{R}$ where $V_i = \varphi(U_i) \subset \mathbb{R}^n$ and we want to make sure g_1 and g_2 are compatible in some sense. We will work extensively with C^∞ functions and so we want that if g_1 is C^∞ , then g_2 better also be C^∞ . These two functions can be related by $g_2 = g_1 \circ \varphi_1 \circ \varphi_2^{-1}$. The function $\varphi_1 \circ \varphi_2^{-1}$, called a transition function, is defined on $V_2 \cap \varphi_2 \circ \varphi_1^{-1}(V_1) \subset \mathbb{R}^n$ (why?). The easiest way to ensure that if g_2 is C^∞ if g_1 is also C^∞ is to declare that $\varphi_1 \circ \varphi_2^{-1}$ is C^∞ . This may seem like too loose of a condition since we may require more out of our transition function to make our answers "agree." It may also seem too restrictive of a condition as there may be looser conditions on the transition functions which still make our answers "agree." However, as we shall soon see, the condition that our transition functions are C^∞ will suffice for our needs, and most maps that we work with are C^∞ anyway, so it isn't needed to find a looser condition.

We now finally arrive at our definition:

Definition 8 (Manifold). *A manifold M is a topological space with open subsets U_i called charts and functions $\varphi_i : U_i \rightarrow \mathbb{R}^n$ called chart maps such that $M = \bigcup U_i$ and the transition functions $\varphi_i \circ \varphi_j^{-1}$ is C^∞ if $U_i \cap U_j \neq \emptyset$ wherever the function is defined.*

Usually a manifold has some extra topological "niceness" conditions such as Hausdorff and separable. While these are necessary for the theorems to hold, it doesn't at all help with the intuition so it's omitted. If you feel like you need an extra condition for a theorem, asking the right questions will lead you to the necessary axiom required.

Sometimes, you will be working in different contexts where instead of C^∞ functions, you want to consider continuous, analytic, or maybe only need k -times differentiable functions. Here, we change the definition so that transition functions are analytic or C^k instead of C^∞ . Also sometimes, since $\mathbb{C}$ is similar to $\mathbb{R}$, we could also look at surfaces which are locally homeomorphic to $\mathbb{C}^n$ where the transition functions are holomorphic. While we will not work with any of these objects, it's good to be aware of the flexibility of the definition. There is an important reason why we use C^∞ functions instead of analytic functions, even though the latter are nicer, but this will be made clear later.

Another thing to note is that from the definition, n could technically vary for each chart map φ_i , but while it doesn't follow directly from the definition, it can be shown that for a connected manifold, the n must be the same for each chart map. This is the sense in which it makes sense to say statements like S^n is n -dimensional.

Exercise (S^1 as a manifold)

Show that S^1 cannot be made homeomorphic to an open subset of $\mathbb{R}^1$, but you can find two open sets $U_1, U_2 \subset S^1$ so that $S^1 = U_1 \cup U_2$ and that there are maps $\varphi_i : U_i \rightarrow \mathbb{R}^1$ which are homeomorphisms to their image. Verify that $\varphi_1 \circ \varphi_2^{-1}$ is C^∞ on the domain of definition (if it isn't, find open sets and chart maps that are).

This exercise makes it clear why we should look locally: we cannot do one-dimensional calculus on S^1 by embedding it into $\mathbb{R}^1$, but we can understand it locally and patch together our answers. Another thing to note is our charts don't necessarily have to be small open sets. In fact, you can choose each U_i above to be the entirety of S^1 minus a point (try it).

We have examples of manifolds like S^n and T^n . It is also useful to look at things which aren't manifolds. One important example is $\mathbb{Q}$ with the induced topology as a subset of $\mathbb{R}$. This is not a manifold as we cannot have a homeomorphism from any open subset of $\mathbb{Q}$ to $\mathbb{R}^n$ for any n (try to prove it). This matches our intuition as we often think about $\mathbb{Q}$ as a sort-of "discrete" object while manifolds should be thought of as "smooth" objects.

3.1 Maps Between Manifolds

As manifolds are topological spaces with extra structure (locally Euclidean), the maps between two manifolds should also have extra structure to reflect this. Essentially, we want the maps to not only

be local homeomorphisms, but also locally differentiable. To formalize this, we make the following definition:

Definition 9 (Differentiable maps). *Given two manifolds M, N , a differentiable map $\varphi : M \rightarrow N$ is a map which is locally C^∞ . More precisely, if we take an open set $V \subset N$ which is homeomorphic to an open subset $V' \subset \mathbb{R}^n$ by $\phi_2 : V \rightarrow V'$ and let $U \subset \varphi^{-1}(V)$ be an open set of M homeomorphic to an open subset $U' \subset \mathbb{R}^m$ via $\phi_1 : U \rightarrow U'$. Then, we define φ to be differentiable if $\phi_2 \circ \varphi \circ \phi_1^{-1} : U' \rightarrow V'$ is C^∞ for all choices of U, V . If φ is in addition a global homeomorphism, it is called a diffeomorphism.*

The main motivation for this definition will come later when we discuss the pullback. The main thing we want is given a C^∞ function $f : N \rightarrow \mathbb{R}$ and a differentiable map $\Phi : M \rightarrow N$, we want that $f \circ \Phi : M \rightarrow \mathbb{R}$ to also be C^∞ . You should think about what's required here and you will find that a differentiable map is exactly what we need.

4 Pullbacks

One thing to observe is that in our first motivating examples, we only got a sense of the topology by looking at the dimension of our cohomology spaces. However, we have that $H^n(X)$ is a vector space, and vector spaces have a lot of structure and the dimension is a fairly weak aspect of the powerful theory of linear algebra. Indeed, historically in the development of homology and cohomology, the only invariant that was studied was the dimension of these vector spaces, which were called the betti numbers of the manifold. Later, it was discovered that studying these invariants as vector spaces is much more powerful, as it allows you to relate the cohomology of spaces which are related. Our first example of this phenomenon will be functoriality which states that if we have a differentiable map between manifolds $\varphi : X \rightarrow Y$, we have an induced map φ^* called the pullback of φ between cohomologies: $\varphi_n^* : H^n(Y) \rightarrow H^n(X)$. The reason that the spaces are reversed will become clear soon. Hence, if we have some function to relate X and Y , we can get some nontrivial relationship between their cohomologies, which give topological information.

For now, we will focus on the condition $X \subseteq \mathbb{R}^{m_1}$ and $Y \subseteq \mathbb{R}^{m_2}$. Given a differentiable map $\varphi : X \rightarrow Y$, to construct the function $\varphi_n^* : H^n(Y) \rightarrow H^n(X)$, we will construct functions (with some arguable abuse of notation) $\varphi_n^* : \Omega^n(Y) \rightarrow \Omega^n(X)$ and show that these induce maps on the cohomology. Let's first construct the map $\varphi_0^* : \Omega^0(Y) \rightarrow \Omega^0(X)$. Ω^0 is just the C^∞ functions on any space, so given a C^∞ function on Y , we want to get a C^∞ function on X . Letting f be a C^∞ function on Y , we can define $\varphi_0^*(f) = f \circ \varphi$ which is a C^∞ function on X (look back at the definition of a differentiable map). The following diagram is a good picture to keep in mind. It says f is a map from Y to $\mathbb{R}$ and if we have a map φ from X to Y , we end up with a map X to $\mathbb{R}$.

$$X \xrightarrow{\varphi} Y \xrightarrow{f} \mathbb{R}$$

$f \circ \varphi$

There is a strong analogy here to dual vector spaces. Suppose we have a linear transformation $T : V \rightarrow W$. We can consider the dual space V^* to be the vector space of linear maps from V to the base field, so pre-composing with T gives a linear map from W to the base field i.e. an element of W^* . Hence, we get a map $V^* \rightarrow W^*$. Looking this way, we can consider the following alternative viewpoint:

Exercise (Pullbacks)

Show that a C^∞ function on Y is equivalent to a differentiable map from Y to $\mathbb{R}$ considered as a manifold in the obvious way. Show also that compositions of differentiable maps is again a differentiable map. Hence interpret the pullback as using a differentiable map from $Y \rightarrow \mathbb{R}$ to get a differentiable map $X \rightarrow \mathbb{R}$ using a differentiable map $\varphi : X \rightarrow Y$.

Now that we have defined φ_0^* , we want to inductively define φ_n^* if we know φ_{n-1}^* . Note, the only way that we know how to go from $n-1$ forms to n forms is using the exterior derivative. We will denote the exterior derivative on Y by d_Y and the exterior derivative on X by d_X . Morally, one way we can get a map $\Omega^n(Y) \rightarrow \Omega^n(X)$ is to go back to $\Omega^{n-1}(Y)$ using d_Y^{-1} (if it were to exist), use our previously constructed φ_{n-1}^* , and use d_X to go to $\Omega^n(X)$. Hence, we want $\varphi_n^* = d_X \circ \varphi_{n-1}^* \circ d_Y^{-1}$. Now, d_Y is not an isomorphism, so the inverse doesn't exist, but the next best thing is to declare $\varphi_n^* \circ d_Y = d_X \circ \varphi_{n-1}^*$.

Note that this is saying that there are two ways to get from $\Omega^{n-1}(Y)$ to $\Omega^n(X)$, and the answer you get from these two paths will match for any $n-1$ -form on Y . Since we want to think of the φ_n^* as "one object" similar to how we think of d , we will often omit indices, so the condition is written more succinctly as $\varphi^* \circ d_Y = d_X \circ \varphi^*$ (sometimes, the X and Y subscripts are also omitted). If this property holds, the diagram is said to commute. This means that the maps d and φ^* commute, or alternatively, between any two spaces connected by a sequence of maps, the answer you get will be independent of the path taken (verify this).

This gives us a nice condition on φ , but doesn't actually tell us what it looks like, or if it even exists. Using this condition however, we can reach a natural definition for this map. Let's look at what φ_1^* should look like. Any 1-form on Y is of the form $\sum_{i=1}^{m_2} g_i dy_i$. Let's first look at how dy_i transforms. Note that $dy_i = d_Y(y_i)$ where y_i is the function on $Y = \mathbb{R}^{m_2}$ which returns the i -th coordinate of each point. Now, we have that $\varphi_1^*(d_Y(y_i)) = d_X(\varphi_0^*(y_i)) = d_X(y_i \circ \varphi)$. Notice that $y_i \circ \varphi$ is a function on X so taking d_X makes sense. We define $\varphi_i = y_i \circ \varphi$ so that $\varphi(x_1, \dots, x_{m_1}) = (\varphi_1(x_1, \dots, x_{m_1}), \dots, \varphi_{m_2}(x_1, \dots, x_{m_1}))$. With this notation, we find $\varphi_1^*(dy_i) = d_X(\varphi_i)$, or written more suggestively, $\varphi_1^*(dy_i) = d\varphi_i$.

Now, for a general 1-form $\sum g_i dy_i$, the only logical thing to send it to is $\varphi_1^*(\sum g_i dy_i) = \sum (g_i \circ \varphi) d\varphi_i$. In general, we define the pullback as follows:

Definition 10 (Pullback). Given a differentiable map $\varphi : Y \rightarrow X$ where X, Y are open subsets of Euclidean space, we define a sequence of functions φ^* called the pullback of φ as $\varphi_n^*(\sum g_I dy_I) = \sum (g_I \circ \varphi) d\varphi_I$ where $d\varphi_i = d(y_i \circ \varphi)$ and $d\varphi_{(i_1, \dots, i_n)}$ is defined as $d\varphi_{i_1} \dots d\varphi_{i_n}$.

Exercise (Type Validation)

Show that the definition above makes sense in terms of everything being in the right space. Confirm by explicit calculation that $d_X \circ \varphi^* = \varphi^* \circ d_Y$.

There are two useful ways of thinking about pullbacks. One is if φ is an embedding, then the pullback tells us how the differential form should look like restricted to the image of X . A second useful point is that if φ is a diffeomorphism (in particular X and Y are of the same dimension), the pullback serves as a change of coordinates.

Exercise (Pullback of an Embedding)

Let $X = \mathbb{R}^1$ with coordinate t and $Y = \mathbb{R}^2$ with coordinates x, y and take a differentiable map $\varphi : X \rightarrow Y$. Describe $\varphi_0^*, \varphi_1^*, \varphi_2^*$ with explicit formulas in terms of $\varphi = (\varphi_1(t), \varphi_2(t))$. If φ is a unit-speed map ($|\varphi'(t)| = 1$ for all t), give a geometric interpretation of the pullback. Then, give an intuitive explanation for the scaling involved when φ is not unit-speed.

Exercise (Jacobian)

Let $X = Y = \mathbb{R}^n$ with X having local coordinates x_i and Y having local coordinates y_i . Consider the n -form $\omega = dy_1 \dots dy_n$. Let the map $\varphi = (y_1(x_1, \dots, x_n), \dots, y_n(x_1, \dots, x_n))$ (this is a slight abuse of notation as we are using y_i to represent both the coordinates and the functions, but there's a good reason). Show that the pull back of this form is $\varphi^*(\omega) = \det \left(\frac{\partial y_i}{\partial x_j} \right) dx_1 \dots dx_n$. Interpret this result geometrically (hint: an n -form that looks like $dx_1 \dots dx_n$ is called a volume form, think of what it represents in an integral). Interpret the sign at each point geometrically.

Exercise (Round Coordinates)

Let X be the right half space of $\mathbb{R}^2$ with coordinates r, θ (note that these aren't polar coordinates, we're just saying the horizontal coordinate is called r and the vertical coordinate is called θ). There is a differentiable map from X to $Y = \mathbb{R}^2 - \{0\}$ given by $(r, \theta) \rightarrow (r \cos(\theta), r \sin(\theta))$. Explicitly compute $\varphi_0^*, \varphi_1^*, \varphi_2^*$.

Do a similar procedure in $\mathbb{R}^3$ with cylindrical and polar coordinates. Do the same for $X = Y = \mathbb{R}^n$ with a linear change of coordinates $\varphi \in GL(n)$.

5 Differential Cochain Complex on Manifolds

Now that we have the language of manifolds, we should go ahead and define the differential cochain complex on a manifold. We saw that C^∞ functions on a manifold X , $f : X \rightarrow \mathbb{R}$ can be described as a collection of C^∞ maps $g_i = f \circ \varphi_i^{-1} : V_i \rightarrow \mathbb{R}$ where $V_i = \varphi_i(U_i)$. We saw that these functions are related in the following manner: $g_j = g_i \circ (\varphi_i \circ \varphi_j^{-1})$. It turns out this condition is enough to understand functions on our manifold. For notational convenience, we will denote the transition functions $\varphi_i \circ \varphi_j^{-1}$ as T_{ij} .

Exercise (Locally Defined Functions)

Show that for a manifold X with charts U_i and chart maps φ_i , a sequence of C^∞ functions g_i defined on $\varphi_i(U_i)$ satisfying the condition $g_j = g_i \circ T_{ij}$ when $U_i \cap U_j \neq \emptyset$, there is a function $f : X \rightarrow \mathbb{R}$ such that $f \circ \varphi_i^{-1} = g_i$. In other words, locally defined functions which agree on intersections define a globally defined function. This exercise shows that instead of first defining a globally defined function on X and then checking that its C^∞ , we can instead locally define C^∞ functions locally and "glue" them together. In a similar fashion, we will define differentiable k -forms on a manifold by first defining them locally on $\mathbb{R}^n$ where we understand them and then "gluing" them together if they're compatible.

We'll use the same philosophy to construct k -forms on a general manifold. Namely, we will define a k -form as a collection of k -forms ω_i defined on $\varphi_i(U_i)$ using the appropriate local coordinates which agree on the intersections of these open sets. If we have these local objects that agree on intersections, we can use them to construct a global object similar to the exercise.

What should we mean by ω_i and ω_j agreeing on $U_i \cap U_j$? As explained in the previous section, the pullback of a differential form under some map $X \rightarrow Y$ shows how the differential form Y "looks like" when restricted to the image of X . In particular, if the map is a diffeomorphism, it represents a sort of change of coordinates. Here, the transition maps (which are diffeomorphisms where defined) allow us to compare differential forms on two different open sets. Hence, our condition on our locally defined differential forms is that $\omega_i = T_{ij}^*(\omega_j)$, which exactly states that the differential forms look the same on intersections based on the change of variables formula.

Definition 11 (Differential Forms (Manifolds)). Given a manifold M with open sets U_i , chart functions $\varphi_i : U_i \cong V_i \subseteq \mathbb{R}^n$, and transition functions $T_{ij} = \varphi_i \circ \varphi_j^{-1}$, we define $\Omega^k(M)$ to be the vector spaces generated by global differential k -forms which are defined with the following data: on each V_i , define a k -form ω_i using the local coordinates on V_i which agree on intersections in the sense that if $U_i \cap U_j \neq \emptyset$, $\omega_i = T_{ij}^*(\omega_j)$.

The following exercises will hopefully get you acquainted with this idea.

Exercise (One Forms on S^1)

One forms intuitively represent vector fields. A vector field on S^1 should intuitively be a vector on each point of the unit circle which is tangent to the circle at that point. Consider such a vector field such that every vector is the same length. Using your decomposition of $S^1 = U \cup V$, write down what this vector field should look like in local coordinates for each open set and see that the local function on one open set agrees with the pullback of

the other. Choose a different homeomorphism for your chart map on U (e.g. scale your current map by 2) and do the same.

Exercise (Exterior Derivative on Manifolds)

Consider a global differential k -form on M given by some local forms $\{\omega_i\}$. In the local coordinates, compute $d\omega_i$ for all i . Show that $d\omega_i$ and $d\omega_j$ agree on the intersection in the sense that $d\omega_i = T_{ij}^*(d\omega_j)$. Conclude there are well defined maps $d_M^k : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$. Show that $d_M^2 = 0$. Now, we can define cohomology on manifolds! It is simply $H^k(M) = \ker(d_M)/\text{im}(d_M)$, where we choose the appropriate indices for the exterior derivative.

Exercise (Pullback on Manifolds)

Let φ be a differentiable map between two manifolds M, N . If we have a k -form on N , we want to obtain a k -form on M to extend our notion of pullback to manifolds. We do this in two steps. First, suppose $M \subset \mathbb{R}^m$ is an open subset. Take a decomposition V_i of charts on N and let ω be a k -form locally defined on these charts. Let $\phi_i : V_i \rightarrow V'_i \subset \mathbb{R}^n$ be chart maps. Pull back the differential form on V'_i to M from the function $\phi_i \circ \varphi$. Show that these pullbacks are compatible, and hence can be glued to a global k -form on M .

Now, for M a general manifold, take an open cover of charts U_i . Let $\psi_i : U_i \rightarrow U'_i \subset \mathbb{R}^m$ be local charts. Define a pullback on each U'_i using $\varphi \circ \psi_i^{-1}$. Show that these agree on intersections, and hence can be glued to a global k -form.

Show that the pullback on differential forms also induces a pullback on the cohomology. Hint: show this pullback commutes with the exterior derivative of the last example.

Exercise (Example of Pullback)

Consider the map $\varphi : S^1 \rightarrow \mathbb{R}^2 - \{0\}$ which embeds S^1 to the unit circle. Look back at the explicit formula for the generator of $H^1(\mathbb{R}^2 - \{0\})$, call it F , and compute the pullback of this 1-form. Note that we know that since F is a closed form, the pullback is also closed and hence $\varphi^*(F)$ represents some cohomology class on $H^1(S^1)$. Unfortunately, this does not tell us $H^1(S^1)$ is non-trivial since it could be true a priori that $\varphi^*(F) = df$ for some $f \in \Omega^0(S^1)$, and we have nothing to tell us otherwise.

Now, consider the map $\psi : \mathbb{R}^2 - \{0\} \rightarrow S^1$ which in polar coordinates sends (r, θ) to θ (in other words, it projects the punctured plane onto the unit circle). What is $\psi^*(\varphi^*(F))$? Consider this as a map $H^1(S^1) \rightarrow H^1(\mathbb{R}^2 - \{0\})$ and use it. Note: this suggestive answer does not follow from functoriality, and is instead due to the special nature of the function we chose.

We have now used functoriality to conclude that $H^1(S^1)$ was non-trivial. This was to be expected as S^1 has a "one-dimensional hole." This shows the power of the cohomology vector spaces further than just the dimension (i.e. Betti numbers). By relating S^1 to $\mathbb{R}^2 - \{0\}$ topologically, we were able to relate the cohomology spaces algebraically. Since linear algebra is much easier to understand than topology, we were able to utilize the algebraic information associated to a space we understand to conclude topological information of a space we don't (concluding that S^1 has a hole).

However, we merely showed that $H^1(S^1)$ is non-trivial, but we don't know that it doesn't have dimension higher than one. If the dimension was higher, it would imply S^1 had multiple 1-dimensional holes, which doesn't make sense intuitively. As an exercise, use functoriality to conclude it must be 1-dimensional. Nevertheless, the fact that we understood $\mathbb{R}^2 - \{0\}$ and had an embedding of S^1 into this space was very coincidental. In general, we do not have such power to embed a manifold M into a manifold we understand and in a manner which is nice on cohomology. We usually only have access to local information, and this last part of the talks will outline how we can use an understanding of the cohomology of subsets to deduce the cohomology of the entire space.

The theorem yields what is called the Mayer-Vietoris sequence and proceeds roughly as follows: suppose M is the union of two open subsets U, V which we understand (for example, S^1 is the union of two open intervals, namely the charts), then we can glue together the local cohomological information

to deduce the cohomology of M . As suggested when trying to define k -forms on a manifold, we need to understand not only the cohomology of U and V , but also the cohomology of their intersection $U \cap V$. For example, the disjoint union of two intervals and S^1 can both be expressed as the union of two intervals, but the intersection of the intervals looks very different in these two spaces.

6 Partitions of Unity

One crucial tool that we will need in what follows is a method to be able to glue together local information. What was outlined in the exercises of the previous notes was that if we can locally define a k -form, then it glues to a global k -form, and we can define this global k -form as taking the values at some point of the local k -form of any open set that contains that point. We will need a more convenient tool for gluing together these local forms, and this is what we call a partition of unity. Partition of unity is actually a pretty good name for what it actually is: you're splitting up one, or more precisely, you're dividing the constant function 1 into the open sets of M .

Definition 12 (Partition of Unity). *When we have a set of open charts $M = \bigcup U_\alpha$, a partition of unity subordinate to U_α is a set of C^∞ functions ρ_α on M such that:*

- *The support of ρ_α lies inside U_α . The support of a function is the closure of the set on which the function attains a nonzero value. Intuitively, this means the function is 0 outside of U_α*
- $\sum_\alpha \rho_\alpha(p) = 1$ for all $p \in M$.

If we relax the C^∞ condition, the following is an example of a partition of unity on $\mathbb{R}$ which can be considered as the union of $U_1 = (-\infty, 2)$ and $U_2 = (-1, \infty)$.

$$\rho_1(x) = \begin{cases} 1 & x \leq 0 \\ 1-x & 0 \leq x \leq 1 \\ 0 & 1 \leq x \end{cases}$$

$$\rho_2(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 \leq x \leq 1 \\ 1 & 1 \leq x \end{cases}$$

Verify this does indeed satisfy the axioms above. The reason we want C^∞ functions is apparent when we look at how we use these: given local k -forms ω_α , each ω_α is a priori only defined on U_α , but $\omega_\alpha \rho_\alpha$ can be interpreted as a k -form defined on all of M (since ρ_α is 0 outside U_α). Then, we can define a global k -form $\omega = \sum_\alpha \omega_\alpha \rho_\alpha$ which agrees with every ω_α when restricted to U_α . Verify this last claim using the definitions of a global k -form and a partition of unity! The reason we want ρ_α to be C^∞ is because we will want to consider forms of the form $\tau \rho_\alpha$ (for some form τ) for its own right and use it to compute cohomology. In particular, we want to ensure that we can take arbitrarily many derivatives of this object. Since τ already consists of C^∞ functions (since it's already a form), all we need is ρ_α is C^∞ .

Hopefully this at least partially motivates why such an object is useful, but the issue is there's no clear way on how to find such a C^∞ partition of unity, or if one even exists. Note: the following construction is where a lot of the technical conditions of a manifold come from. I will only provide a sketch so I will omit a lot of details, but you may be able to notice some of the flaws and try to fix them. If so, check the rigorous definition of a manifold with a source and see if your solution was right. Something else to note is that this is why we work with C^∞ functions instead of analytic functions, even though the latter are often nicer: it turns out we can't find an analytic partition of unity! Try to think about why (maybe after reading the construction).

6.1 Sketch of the Construction of a C^∞ Partition of Unity

In order to construct such an object, we will make a lot of simplifications and then just find an explicit function that does what we want. The first simplification we can make is that we can relax the condition that $\sum \rho_\alpha = 1$ and replace it with the condition that $\sum \rho_\alpha$ is nonzero everywhere. The reason is if $f = \sum \rho_\alpha$ and $f(p) \neq 0$ for all $p \in M$, then $\{\frac{\rho_\alpha}{f}\}$ is a well defined C^∞ function and sums to 1 everywhere, hence it is a partition of unity.

Now we just need C^∞ functions on U_α such that at every point, the sum is nonzero, so for convenience we can restrict to all functions being non-negative and say that we just need one function at every point to be nonzero. Now, consider the chart maps $\varphi_\alpha : U_\alpha \rightarrow V_\alpha \subset \mathbb{R}^n$. We can equivalently find non-negative C^∞ functions on V_α whose support is contained in V_α (since V_α is open, this intuitively equates to the function vanishing as it approaches the boundary). Since V_α is open, we can cover it with open balls (or restrict to charts so that U_α is itself homeomorphic to an open ball, which is often taken for the definition of a manifold; this is not as restrictive of a condition as you may initially believe, think about why). Now, we can look for some C^∞ function on a ball, take it to be the unit ball for convenience, which is non-negative and vanishes as it approaches the boundary. We can choose this function to be radially symmetric and hence the problem reduces to the same problem for the interval $(-1, 1)$ instead of a unit ball in $\mathbb{R}^n$ (why?). Now, we just construct such a function. As mentioned above, we can make a C^∞ partition of unity, but not an analytic one. Hence, we will utilize the go-to C^∞ function that isn't analytic:

Exercise (e^{-1/x^2})

Define $f(x) = e^{-1/x^2}$ for $x \neq 0$ and $f(0) = 0$. Verify that f is continuous everywhere. Compute all derivatives of $f(x)$ at 0 (may be useful to work directly from the definition). Compute the Taylor series of f at 0 and show that it does not converge to f except at 0. Conclude that f is C^∞ , but not analytic.

If you graph e^{-1/x^2} , then you will see it gets "very flat" near 0. We will exploit this property: define $g(x)$ to be $f(x)$ for $x \leq 0$ and $g(x) = 0$ for $x \geq 0$. Verify that g is continuous, everywhere non-negative, and C^∞ everywhere. What is its support (the closure of the set where it takes nonzero value).

We want a similar function which goes to 0 at both -1 and 1. For this, defining something like $F(x) = e^{-1/\cos^2(\pi x/2)}$ for $x \in (-1, 1)$ and $F(x) = 0$ outside this interval. Verify that $F(x)$ is C^∞ and non-negative everywhere and has support $[-1, 1]$. This function can be used to construct our partition of unity by reversing the steps. Since this was a sketch, there are some details omitted and some inaccuracies, but the goal was to portray that it is reasonable to be able to construct a C^∞ partition of unity on any manifold.

7 Mayer-Vietoris

Now that we have partitions of unity, let's get into the Mayer-Vietoris sequence. As a reminder of the setup, we have a manifold M which can be decomposed into open sets $M = U_1 \cup U_2$ where we hopefully understand the cohomology of U_i and $U_1 \cap U_2$, and we want to use this information to deduce the cohomology of M . The process is as follows: we start with some topological maps, show that these give us what is called an exact sequence (nice maps on the vector spaces) of spaces of the k -chains using partitions of unity, and finally show this gives us an exact sequence of cohomology. First, we can include $U_1 \cap U_2$ into $U_1 \amalg U_2$ in two ways: ι_1 includes it into the first factor and ι_2 includes it into the second factor. Further, there is a map $\varphi : U_1 \amalg U_2 \rightarrow M$ defined to take a point on the disjoint union to its corresponding point on M . Hence we have a sequence of maps

$$M \xleftarrow{\varphi} U_1 \amalg U_2 \begin{matrix} \xleftarrow{\iota_1} \\ \xleftarrow{\iota_2} \end{matrix} U_1 \cap U_2$$

leading to pullback maps on the differential chains

$$\Omega^k(M) \xrightarrow{\varphi^*} \Omega^k(U_1 \coprod U_2) \xrightarrow[\iota_2^*]{\iota_1^*} \Omega^k(U_1 \cap U_2).$$

We make two modifications to the sequence above. First, $\Omega^k(U_1 \coprod U_2) \cong \Omega^k(U_1) \oplus \Omega^k(U_2)$. Secondly, we take the difference $\iota_1^* - \iota_2^*$ and denote this map as ψ^* (it isn't the pullback of any map, but the asterisk is added for consistency of naming maps). The reason for considering this map will become clear soon. Our sequence now looks like this:

$$\Omega^k(M) \xrightarrow{\varphi^*} \Omega^k(U_1) \oplus \Omega^k(U_2) \xrightarrow{\psi^*} \Omega^k(U_1 \cap U_2).$$

Let's describe these spaces and maps in detail. The first space consists of all global differential forms on M , the second consists of pairs of differential forms with the first defined on U_1 and the second defined on U_2 , and the third consists of differential forms defined on the intersection $U_1 \cap U_2$. Now, φ^* takes a global differential form ω on M to the pair $(\omega|_{U_1}, \omega|_{U_2})$ and ψ^* takes a pair of forms (ω, τ) to the form $\omega|_{U_1 \cap U_2} - \tau|_{U_1 \cap U_2}$ (verify that these descriptions are indeed correct). It can be easily checked that both of these maps commute with the differential map between Ω^k , and hence we get induced maps

$$H^k(M) \xrightarrow{\varphi^*} H^k(U_1) \oplus H^k(U_2) \xrightarrow{\psi^*} H^k(U_1 \cap U_2)$$

as you should be comfortable with by now (verify this statement if not). The magical part of the Mayer-Vietoris sequence is that since the maps φ^* and ψ^* are "nice" (exact), we get a magical map d^* (it is in fact related to the differential map d) from $H^k(U_1 \cap U_2) \rightarrow H^{k+1}(M)$ called the connecting homomorphism making the diagram below exact. Exactness of sequences and why it's a very nice condition will be explained below. Essentially, it means that we can recover a vector space in the sequence if we know the vector spaces around it.

$$\begin{array}{ccccccc} H^{k+2}(M) & \xleftarrow{\quad} & & \xrightarrow{\quad} & \dots & & \\ & & & \searrow d^* & & & \\ H^{k+1}(M) & \xleftarrow{\varphi^*} & H^k(U_1) \oplus H^k(U_2) & \xrightarrow{\psi^*} & H^{k+1}(U_1 \cap U_2) & & \\ & & & \searrow d^* & & & \\ H^k(M) & \xleftarrow{\varphi^*} & H^k(U_1) \oplus H^k(U_2) & \xrightarrow{\psi^*} & H^k(U_1 \cap U_2) & & \\ & & & \searrow d^* & & & \\ & & & & \dots & \xrightarrow{\quad} & H^{k-1}(U_1 \cap U_2) \end{array}$$

Once we describe what d^* is, we will be able to deduce $H^k(S^1)$ and even $H^k(S^n)$ once we prove another intermediate theorem.

7.1 Exact Sequences

Exact sequences are a very useful tool that shows up all across algebra. The goal of this section is to give an introduction to this idea and some intuition behind it.

Definition 13 (Exact Sequence). A sequence of maps $A \xrightarrow{f} B \xrightarrow{g} C$ is said to be exact at B if $\ker(g) = \text{im}(f)$. A sequence of maps $\dots \rightarrow A_{i-1} \xrightarrow{f_{i-1}} A_i \xrightarrow{f_i} A_{i+1} \rightarrow \dots$ is said to be exact if it is exact at every step. In other words, $\ker(f_i) = \text{im}(f_{i-1})$ for all i . An exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is said to be a short exact sequence.

A couple notes: in the last sentence defining a short exact sequence, I omitted the maps. This is often done with exact sequences for notational convenience, but always keep in mind that there are maps in the background which makes the sequence exact. I also haven't stated what the 0 is, or for that matter, what any of the A, B, C or other objects actually are. The versatility of the language of exact sequences is that the objects can be a multitude of things. Since we're talking about kernels and images, they should be some sort of objects where this makes sense such as groups, vector spaces, R -modules, etc. with the maps being homomorphisms, linear maps, etc. Here, the 0 object always represents the "trivial" object e.g. the trivial group, the 0 vector space, the 0 module etc. When discussing the 0 object, there's only one possible map coming from it and one possible map going to it (verify this with the examples) and so these maps from and to 0 are always implied. Note that elsewhere, sometimes when talking about groups multiplicatively, the trivial object is notated as 1 instead of 0.

For example however, the objects and maps cannot be something like topological spaces and continuous maps, because it does not make sense to take a kernel in this scenario. This next example should help see why this is a convenient language to have.

Exercise (Basic Properties of Exact Sequences)

Suppose the sequence $0 \rightarrow A \xrightarrow{f} B$ is exact. Show that f is an injection of A into B . Similarly, suppose the sequence $A \xrightarrow{f} B \rightarrow 0$ is exact. Show that f is a surjection from A to B . Suppose the sequence $0 \rightarrow A \xrightarrow{f} B \rightarrow 0$ is exact. Show that f is an isomorphism from A to B .

Hence, we can nicely encode injections, surjections, and isomorphisms into exact sequences. This is more than just fancy bookkeeping, if we find such a sequence "in the wild" such as if something like $0 \rightarrow A \rightarrow B \rightarrow 0$ shows up in a Mayer-Vietoris sequence, we can use this to conclude that $A \cong B$. Another useful thing to intuit is the short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$. We should interpret a short exact sequence as " C is the quotient B/A ." Check that this statement makes sense in the context of groups and vector spaces. Vector spaces, which is where we will be using this language, are actually a very nice context in which to use exact sequences because a short exact sequence can be equivalently stated as $B \cong A \oplus C$. This is not true in general, for example the analog in groups would be that B is the direct product of A and C . But if we have a short exact sequence of groups, we can also have something like B is some sort of semi-direct product of A and C (find some example of this). Short exact sequences are important because they essentially are the building blocks for arbitrary exact sequences.

Exercise (Decomposing to Short Exact Sequences)

Suppose we have an exact sequence $\dots \rightarrow A_{i-1} \xrightarrow{f_{i-1}} A_i \xrightarrow{f_i} A_{i+1} \rightarrow \dots$. Show that we can extract short exact sequences $0 \rightarrow \text{im}(f_{i-1}) \rightarrow A_i \rightarrow \ker(f_i) \rightarrow 0$. To see how everything fits together, show that given our initial exact sequence, we can construct the exact sequence

$$\dots \rightarrow A_{i-1} \rightarrow \ker(f_{i-1}) \rightarrow 0 \rightarrow \text{im}(f_{i-1}) \rightarrow A_i \rightarrow \ker(f_i) \rightarrow 0 \rightarrow \text{im}(f_i) \rightarrow A_{i+1} \rightarrow \dots$$

Exercise (Short Exact Sequences of Vector Spaces)

This problem will demonstrate the usefulness of using exact sequences to determine information about vector spaces. If $0 \rightarrow \mathbb{R}^n \rightarrow V \rightarrow \mathbb{R}^m \rightarrow 0$ is an exact sequence of vector spaces, compute the dimension of V ? If $0 \rightarrow \mathbb{R}^k \rightarrow \mathbb{R}^m \rightarrow V \rightarrow \mathbb{R}^n \rightarrow 0$ is exact, compute the dimension of V .

The way we will utilize this tool is we will get some maps with the unknown vector space as some $H^k(M)$ and it will be surrounded by spaces $H^k(U \cap V)$ and $H^k(U) \oplus H^k(V)$. From these examples, we see that we should be able to deduce the dimension of $H^k(M)$. In fact, once we construct the maps d^* , we will be able to construct explicit generators for $H^k(M)$ as well.

7.1.1 Exact Sequences of Cochain Complexes

We will utilize the language of exact sequences to relate the cohomology vector spaces, in which case we work with exact sequences of cochain complexes. The objects here are cochain complexes. A cochain complex is a sequence of vector spaces Ω^k indexed by nonnegative integers with maps $d^k : \Omega^k \rightarrow \Omega^{k+1}$. Hence, the collection of differential k -forms with maps $\Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \dots$ is a chain complex and will be denoted $\Omega^*(M)$. Note that d is not a map between chain complexes, but is encoded into the data of one chain complex. A map between chain complexes $\Omega^*(X) \rightarrow \Omega^*(Y)$, called a chain map, is a collection of linear maps $\varphi_k : \Omega^k(X) \rightarrow \Omega^k(Y)$ which commutes with d i.e. such that $\varphi \circ d = d \circ \varphi$ where the first d refers to the map on $\Omega^*(X)$ and the second on $\Omega^*(Y)$. We should be very comfortable with these maps by now, and the reason such maps are useful is that they induce maps on the cohomology of the cochain complexes.

Exercise (Exact Sequences of Cochain Complexes)

Show that the map $\varphi^* : \Omega^*(M) \rightarrow \Omega^*(U_1) \oplus \Omega^*(U_2)$ is a map of cochain complexes (what does the direct sum of two chain complexes mean?). and $\psi^* : \Omega^*(U_1) \oplus \Omega^*(U_2) \rightarrow \Omega^*(U_1 \cap U_2)$ is also a map of cochain complexes. What should the 0 object be for cochain complexes (determine what each k -indexed vector space is and the maps d between them). Show that $0 \rightarrow \Omega^*(M) \xrightarrow{\varphi^*} \Omega^k(U_1) \oplus \Omega^k(U_2) \xrightarrow{\psi^*} \Omega^k(U_1 \cap U_2)$ is exact. That is, show that φ^* is injective and that $\ker(\psi^*) = \text{im}(\varphi^*)$. Essentially, you need to show that exactness holds for each map on Ω^k .

We now use the power of partitions of unity to show that in fact, ψ^* is surjective, or in other words, given a differential form $\tau \in \Omega^k(U_1 \cap U_2)$, there is a pair of differential forms $\omega_i \in \Omega^k(U_i)$ such that $\omega_1|_{U_1 \cap U_2} - \omega_2|_{U_1 \cap U_2} = \tau$. Hopefully this result should be at least a bit surprising. The upshot is that we can make our exact sequence into a short exact sequence of cochain complexes $0 \rightarrow \Omega^*(M) \xrightarrow{\varphi^*} \Omega^k(U_1) \oplus \Omega^k(U_2) \xrightarrow{\psi^*} \Omega^k(U_1 \cap U_2) \rightarrow 0$. This short exact sequence will be crucial in what follows.

In order to show ψ^* is surjective, we must construct our desired pair (ω_1, ω_2) given a form τ on $U_1 \cap U_2$. Let ρ_1, ρ_2 be a partition of unity subordinate to U_i . When we used them to construct a global form from a local form, we looked at forms of the form $\omega \rho_i$. Note that here $\tau \rho_1$ isn't a form on U_1 since ρ_1 can have a nonzero value on $U_1 - U_2$, and it isn't clear how to define $\tau \rho_1$. However, we have the remarkable property that $\tau \rho_1$ can be interpreted as a form on U_2 ! This is because ρ_1 is 0 on $U_2 - U_1$ and $\tau \rho_1$ is well defined on $U_1 \cap U_2$ (make sure you're comfortable with this, the switching of indices is important). Similarly, we can define $\tau \rho_2$ to be a form on U_1 . Hence, we can let $(\omega_1, \omega_2) = (\tau \rho_2, -\tau \rho_1)$ so that when restricted to $U_1 \cap U_2$, $\omega_1 - \omega_2 = \tau(\rho_2 - (-\rho_1)) = \tau$.

8 The Mayer-Vietoris Sequence

We will utilize the short exact sequence $0 \rightarrow \Omega^*(M) \rightarrow \Omega^*(U_1) \oplus \Omega^*(U_2) \rightarrow \Omega^k(U_1 \cap U_2) \rightarrow 0$ to construct the "long exact sequence"

$$0 \rightarrow H^0(M) \xrightarrow{\varphi^*} H^0(U_1) \oplus H^0(U_2) \xrightarrow{\psi^*} H^0(U_1 \cap U_2) \xrightarrow{d^*} H^1(M) \rightarrow \dots$$

where the diagram proceeds as described earlier. There are two ways to look at this sequence. The first is to derive it in a purely algebraic manner by formally exploiting the properties of the exact sequence of cochain complexes. This plays into a common motto in algebraic topology: short exact sequences of (co)chain complexes give long exact sequences of (co)homology. However, since we explicitly understand what the maps φ^* and ψ^* are, we can come up with an explicit formula for d^* as well and this will allow us to get a sense of the generators for $H^k(M)$. We will see that the map d^* will be strongly dependent on a choice of partition of unity, but since cohomology should be independent of this choice, we should have that the differential forms constructed give the same answer in cohomology and hence should differ by an exact form.

8.1 Purely Algebraic Viewpoint

Let's start with the algebraic description.

Exercise (Exactness in Cohomology)

Show that the induced maps φ^* and ψ^* on cohomology are exact at $H^k(U_1) \oplus H^k(U_2)$ (use that they are exact on cochains). Also show that $\ker(\varphi_0^*) = 0$ so that the first map of our long exact sequence is injective and we can add a 0 vector space to represent this. You can try to show that $\ker(\varphi_k^*) = 0$ or $\text{im}(\psi_k^*) = 0$, but you will be unsuccessful. The long exact sequence fixes this failure of exactness.

Now, we must construct our connecting homomorphism d^* . Let's look at part of our exact sequence of cochain complexes:

$$\begin{array}{ccccccc}
 & \dots & \uparrow d & & \dots & \uparrow d & & \dots & \uparrow d \\
 0 & \longrightarrow & \Omega^{k+1}(M) & \xrightarrow{\varphi^*} & \Omega^{k+1}(U_1) \oplus \Omega^{k+1}(U_2) & \xrightarrow{\psi^*} & \Omega^{k+1}(U_1 \cap U_2) & \longrightarrow & 0 \\
 & & \uparrow d & & \uparrow d & & \uparrow d & & \\
 0 & \longrightarrow & \Omega^k(M) & \xrightarrow{\varphi^*} & \Omega^k(U_1) \oplus \Omega^k(U_2) & \xrightarrow{\psi^*} & \Omega^k(U_1 \cap U_2) & \longrightarrow & 0 \\
 & & \uparrow d & & \uparrow d & & \uparrow d & & \\
 & \dots & & & \dots & & \dots & &
 \end{array}$$

What we want to do is start with a k -form $\tau \in \Omega^k(U_1 \cap U_2)$ and end up with a $k+1$ form $\omega \in \Omega^{k+1}(M)$. First, we will describe the procedure purely algebraically and then go back and do the same using the explicit descriptions of the maps. I recommend reading over both sections multiple times in succession to internalize how they interact.

Let's start with some cohomology class $[\tau] \in H^k(U_1 \cap U_2)$. From the definition of cohomology, this is the image of some closed form $\tau \in \Omega^k(U_1 \cap U_2)$. By the exactness of the diagram, ψ^* is surjective so there is some $\xi \in \Omega^k(U_1) \oplus \Omega^k(U_2)$ such that $\psi^*(\xi) = \tau$. ξ represents a pair of forms, but we wrote it as one thing to outline that the argument doesn't need this. Now, we have that $d(\xi) \in \Omega^{k+1}(U_1) \oplus \Omega^{k+1}(U_2)$. We're going to use exactness again to get a form $\omega \in \Omega^{k+1}(M)$. Let's denote $d(\xi)$ as ξ' and look at $\psi^*(\xi')$. By commutativity of the diagram (i.e. the definition of a chain map), we have that $\psi^*(\xi') = \psi^*(d(\xi)) = d(\psi^*(\xi)) = d(\tau) = 0$ where in the last step, we used τ is closed since it's a representative for an element of cohomology. Hence, $\xi' \in \ker(\psi^*)$ and by exactness, we must have that $\xi' \in \text{im}(\varphi^*)$. In other words, there is some $\omega \in \Omega^{k+1}(M)$ such that $\varphi^*(\omega) = \xi'$. This is the ω that we desire and the map d^* takes the cohomology class $[\tau]$ to the cohomology class $[\omega]$.

Exercise (Verifications and Diagram Chasing)

Grab some paper and repeat each of the steps above. Yes, really. The argument won't make sense unless you do it yourself, and this is one of those calculations that everyone should go through at least once in their lives.

There are a couple of verifications that need to be made as well to make sure this map is well defined. First, we need to make sure that ω is actually a closed form so that $[\omega]$ actually makes sense. Show this. Hints: use commutativity and exactness, what's $d\xi'$?

There were a couple of choices we made and we must make sure that our answer in cohomology is independent of these choices. First, we chose a representative τ of a cohomology class $[\tau]$, but we also could have chosen some $\tau + d\zeta$ for $\zeta \in \Omega^{k-1}(U_1 \cap U_2)$. Verify that this choice does not change the value of ω . Next, we had that ψ^* was surjective, but we made a choice of ξ . Show that a different choice again changes our answer by at worst an exact form so that when we quotient to the cohomology, we get the same cohomology class. Both of these are just an exercise in being clever with the exactness of the rows and commutativity of the maps. You may think that we also chose some ω , but in fact, once we knew it existed, it must be unique (why?).

For the truly brave: verify that the long exact sequence described above is actually exact at every step.

Now, let's use this powerful machinery to analyze S^1 . We have that $S^1 = U \cup V$ where U_1 and U_2 are both homeomorphic to open intervals and $U_1 \cap U_2$ is homeomorphic to the disjoint union of two intervals. We have here that $H^k(U_1) \oplus H^k(U_2) \cong H^k(U_1 \cap U_2)$ and both are $\mathbb{R}^2$ for $k = 0$ and 0 otherwise (why?). We have that $H^0(S^1) = \mathbb{R}$ because the 0-th cohomology counts connected components (verify that for any manifold M , $H^0(M) = \mathbb{R}^c$ where c is the number of connected components directly from definitions). Hence, our exact sequence proceeds as

$$0 \rightarrow H^0(S^1) \rightarrow H^0(U_1) \oplus H^0(U_2) \rightarrow H^0(U_1 \cap U_2) \rightarrow H^1(S^1) \rightarrow H^1(U_1) \oplus H^1(U_2) \rightarrow \dots$$

or

$$0 \rightarrow \mathbb{R} \rightarrow \mathbb{R}^2 \rightarrow \mathbb{R}^2 \rightarrow \mathbb{R}^2 \rightarrow H^1(S^1) \rightarrow 0 \rightarrow \dots$$

Exercise ($H^k(S^1)$)

Verify that $H^1(S^1) \cong \mathbb{R}$ and $H^k(S^1) = 0$ for $k > 1$.

This matches our intuition topologically that S^1 has one 1-dimensional hole and nothing else. While knowing the dimension is great, we don't actually know what the vector space looks like. We would like to know an explicit differential form which generates the cohomology. To determine this, let's revisit our steps in constructing d^* , but actually use the maps to see what the resulting form looks like.

8.2 Revisiting the Mayer-Vietoris Sequence

In order to obtain ω from τ , our first task was to obtain a pre-image ξ . You will recall from earlier that we can construct a preimage using a partition of unity ρ_1, ρ_2 as $\xi = (\tau\rho_2, -\tau\rho_1)$. Now, we compute that $\xi' = (d(\tau\rho_2), d(-\tau\rho_1))$ and we have that these two differential forms agree on the intersection $U_1 \cap U_2$ by the formal derivation from above. Hence, we can define ω as $d(\tau\rho_2)$ when our point is in U_1 or as $d(-\tau\rho_1) = -d(\tau\rho_1)$ when our point is in U_2 . Perhaps more succinctly, we can use our partitions of unity to patch these local forms together: $\omega = \rho_1 d(\tau\rho_2) - \rho_2 d(\tau\rho_1)$. Hence, $d^*([\tau]) = [\rho_1 d(\tau\rho_2) - \rho_2 d(\tau\rho_1)]$. This final formula is a very explicit formula for how to get ω from τ , but it has a very strong dependence on the partition of unity chosen.

Let's look at an illustrative example. Let $M = (-2, 2)$ and partition it into $U_1 = (-2, 1)$ and $U_2 = (-1, 2)$. $U_1 \cap U_2 = (-1, 1)$. Let's find a nontrivial function in $\Omega^0(U_1 \cap U_2) = C^\infty(-1, 1)$ which is in the kernel of d and see what it gets sent to under d^* . The functions in $\ker(d^0)$ consist of locally constant functions, which in this case is just some constant function. So let's let $\tau = a$ for some constant a . Take a partition of unity ρ_i subordinate to U_i . Now, $\xi = (a\rho_2, -a\rho_1)$ and based on our formal argument, we should have that $\frac{d}{dx}(a\rho_2) = \frac{d}{dx}(-a\rho_1)$. But this follows directly from the fact that ρ_i forms a partition of unity (why?). Now, since the derivatives agree on the intersections, we can glue the derivatives together to get a 1-form on $(-2, 2)$. If you want, try to find an explicit partition of unity and draw graphs of each step. Note something interesting here: the map d^* required our map d , but we didn't differentiate the differential form τ itself, and instead, we differentiated the partition of unity!

Exercise (Generator for $H^1(S^1)$)

Sketch what a partition of unity on S^1 subordinate to U_1, U_2 as used above would look like based on our earlier construction of ρ_i using e^{-1/x^2} . A generator for the cohomology class of $H^0(U_1 \cap U_2)$ takes on a nonzero constant value on one of the open intervals of the intersection and is 0 on the other open interval. Go through the steps and see what d^ of this form looks like. It should be something like a bump on the open interval of intersection where our initial form was nonzero.*

Notice that this form we obtained from the Mayer-Vietoris sequence is actually quite distinct from the generator of $H^1(S^1)$ we saw from the end of the last set of notes, which was uniform everywhere on the circle.

Mayer-Vietoris is actually quite a powerful tool. With an additional theorem, we can actually compute $H^k(S^n)$ for all $n \geq 0$!

Theorem 8.1 (Poincaré Lemma). *For any manifold M , $H^k(M \times \mathbb{R}) \cong H^k(M)$.*

If we take the topological interpretation of the cohomology, this makes sense since for example, going from a circle to a cylinder doesn't change the number of holes in the way that we have been describing. Note that there is a homeomorphism of $\mathbb{R}^2 - \{0\}$ with $S^1 \times \mathbb{R}$ so in this case, the theorem states that $H^k(S^1) \cong H^k(\mathbb{R}^2 - \{0\})$.

The proof utilizes something called a chain homotopy. We will not write it out every step in detail, but sketch out the proof of this theorem for $M = \mathbb{R}^n$. The general case can be recovered by repeating all of the following arguments locally. We have an inclusion map $\iota : \mathbb{R}^n \rightarrow \mathbb{R}^n \times \mathbb{R}$ which takes $\mathbb{R}^n$ to the "zero section," i.e. $\iota(x_1, \dots, x_n) = (x_1, \dots, x_n, 0)$ and a projection map $\pi : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ defined as $\pi(x_1, \dots, x_n, t) \rightarrow (x_1, \dots, x_n)$. These induce maps ι^*, π^* as depicted below.

$$\begin{array}{ccc} \mathbb{R}^n \times \mathbb{R} & & \Omega^*(\mathbb{R}^n \times \mathbb{R}) \\ \uparrow \iota & & \uparrow \iota^* \\ \mathbb{R}^n & & \Omega^*(\mathbb{R}^n) \\ \downarrow \pi & & \downarrow \pi^* \end{array}$$

Since $\pi \circ \iota$ is the identity, we have that $\iota^* \circ \pi^*$ is also the identity on differential forms, and hence also on cohomology. However, $\iota \circ \pi$ takes $(x_1, \dots, x_n, t)$ to $(x_1, \dots, x_n, 0)$ and so $\pi^* \circ \iota^*$ is not the identity on the level of differential forms. However, it turns out that it is the identity on cohomology. To show this, let 1 denote the identity map on differential forms. We want to show that $(1 - \pi^* \circ \iota^*)(\omega)$ is an exact form for all ω so that it induces the zero map on cohomology and $\pi^* \circ \iota^*$ becomes the identity in the quotient. To do this, we will construct a so-called chain homotopy.

Exercise (Chain Homotopy)

A chain map between cochain complexes $\Omega^*(X)$ and $\Omega^*(Y)$ is a sequence of maps $K : \Omega^k(X) \rightarrow \Omega^{k-1}(Y)$ (what should the map on $\Omega^0(X)$ be?). Show that if $f^*, g^* : \Omega^*(X) \rightarrow \Omega^*(Y)$ are two chain maps and there exists a chain homotopy $K : \Omega^*(X) \rightarrow \Omega^*(Y)$ such that $f^* - g^* = \pm(dK \pm Kd)$, then f^* and g^* induce the same map on cohomology $H^*(X) \rightarrow H^*(Y)$.

We construct such a chain homotopy $K : \Omega^*(\mathbb{R}^n \times \mathbb{R}) \rightarrow \Omega^*(\mathbb{R}^n \times \mathbb{R})$ which will satisfy the property above for $f^* = 1$ and $g^* = \pi^* \circ \iota^*$. To construct this map, note that any k -form on $\mathbb{R}^n \times \mathbb{R}$ are linear combinations of forms of one of two types:

$$1 : \pi^*(\phi)f(x, t)$$

$$2 : \pi^*(\phi)f(x, t)dt$$

where ϕ is a differential form on $\mathbb{R}^n$ (of degree k for forms of type 1 and degree $k - 1$ for forms of type 2). To intuit this for 1-forms, let $\phi_1, \dots, \phi_n$ be a basis for $\mathbb{R}^n$, considered as a constant vector field with the vector at each point pointing in the same direction. Forms of type 1 look like $\pi^*(\phi_i)f(x, t)$ with $f(x, t)$ saying how much the vector at each point on $\mathbb{R}^n \times \mathbb{R}$ points in the i -th direction. This covers all "horizontal" vector fields. Forms of type 2 look like $f(x, t)dt$ which dictates how much each vector points vertically. Hence, any vector field can be decomposed into a sum of $\pi^*(\phi_i)f_i(x, t)$ and $g(x, t)dt$ for suitable f_i, g .

Now, our map K is "integration along fibers." More precisely, it is defined to take all forms of type 1 to 0 and all forms of type 2 to $\pi^*(\phi) \int_0^t f(x, t')dt'$.

Exercise (Poincaré Lemma)

Verify that for any form on $\mathbb{R}^n \times \mathbb{R}$, $1 - \pi^* \circ \iota^* = \pm(dK - Kd)$ where the sign depends on the degree of the differential form. Thus conclude that $\pi^* \circ \iota^*$ becomes the identity on cohomology and thus $H^k(\mathbb{R}^n \times \mathbb{R}) \cong H^k(\mathbb{R}^n)$. Use induction to show $H^k(\mathbb{R}^n)$ is $\mathbb{R}$ for $k = 0$ and 0 otherwise. This makes sense as regardless of the dimension, we have one connected component and no holes.

The details of the calculations can be found in Bott and Tu. Bott and Tu does not give a very good intuition behind chain homotopies however. Hatcher gives a fairly good exposition for motivating chain homotopies in the context of singular homology, and the algebraic machinery which follows can be used in other contexts like we did here. Intuitively, in singular homology, a chain homotopy is what results when trying to mimic a homotopy but using singular chains. As stated above, the statement is valid for $\mathbb{R}^n$ replaced by any manifold M , and we will use this fact in the following exercises.

Exercise ($H^k(S^n)$)

Decompose S^n into two open charts U_1, U_2 such that $U_1 \cap U_2$ is homeomorphic to $S^{n-1} \times \mathbb{R}$ (actually looks like the product of S^{n-1} and an open interval). Use Mayer-Vietoris, the Poincaré lemma, and induction to show $H^k(S^n) = \mathbb{R}$ for $k = 0, n$ and 0 otherwise. This matches our intuition as S^n has 1 connected component, 1 “ n -dimensional” hole, and no other holes. If you think about what the Poincaré lemma gives as the isomorphism of forms, you will see that this process leads to a “bump form” similar to what we found for S^1 as the generator for each highest dimensional cohomology. It also shows some of the power of this algebraic abstraction of topological properties: to compute the cohomology of S^n , you use a long exact sequence which needs the cohomology of S^{n-1} . And this, you can compute using another exact sequence. This trickery and black magic can sometimes lead to very beautiful results that are obtained very algebraically.

Exercise (H^k of a Torus)

Let $T = S^1 \times S^1$. Decompose T into U_1, U_2 so that U_i is homeomorphic to an open cylinder ($S^1 \times \mathbb{R}$) and the intersection is the disjoint union of two open cylinders. Use this decomposition to compute the dimensions of $H^k(T)$. Can you do the same for $(S^1)^n$? Notice that for the examples of tori and spheres, the dimensions of the k -th cohomology and $n - k$ -th cohomology are the same! This symmetry is not a coincidence and is a consequence of a nice theorem called Poincaré duality. It is beyond the scope of these notes, but it states that for compact manifolds, this symmetry will always hold (notice that it doesn't hold for $\mathbb{R}^n$). This is quite surprising when you consider the consequences, interpreting these dimensions as counting the number of holes!